

## Relational Belief

### Relational Belief Prompt

$I_t$ : Role & Belief instruction

$h_t$ : Observation history

$o_t$ : Current observation

### Relational Belief

P1: Unknown  
P2: Werewolf  
P3: Villager ...



## Action with Relational Belief

### Action Prompt

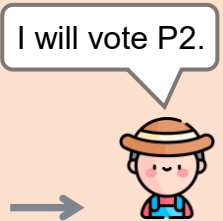
$I_t$ : Role & Action instruction

$h_t$ : Observation history

$o_t$ : Current observation

### Relational Belief

P1: Unknown  
P2: Werewolf ...



(a)

## Speech Action Feedback

✗ P4 is a Werewolf.  
Not sure about others.

✓ P3 is a Werewolf.  
Two days of silence  
is a classic  
Werewolf tell.

P2: Werewolf? →  
**Clearly Werewolf!**



P2: Werewolf! →  
**Maybe not Werewolf.**

## Task Action Feedback

P1: Unknown  
**P2: Werewolf**  
P3: Villager ...

✗ P2 is likely Werewolf.  
But, everyone else votes  
P1, so I will vote for P1.

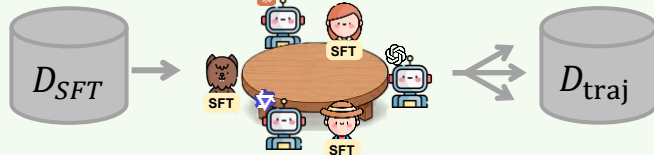


✓ P2 is likely a Werewolf.  
I will vote for P2.

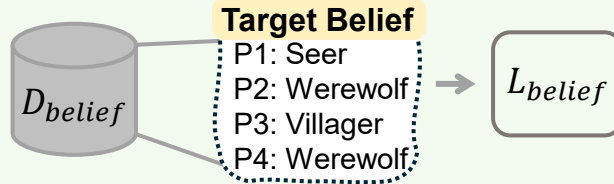
(b)

## Data Construction and Optimization

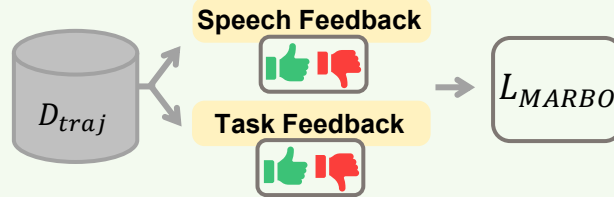
### Trajectory Collection



### Relational Belief Supervision



### Belief-grounded Preference Optimization



(c)